Ameel Aziz
CSCE420
330002274

Homework 5

1a)

P(Pass | Smart, Study) = $\frac{P(Smart, Study, Pass)}{P(Smart, Study)}$

P(pPass | Smart, Study) x P(Smart, Study) = P(Smart, Study, Pass)

Because Smart and Study are independent:

P(Pass | Smart, Study) x P(Smart) x P(Study) = P(Smart, Study, Pass)

1b)

	pass		¬pass	
	¬smart	smart	¬smart	smart
¬study	0.084	0.126	0.336	0.054
study	0.168	0.114	0.112	0.006

1c)
P(smart | pass, -study) = P (smart, pass, -study) / P(pass, -study) = 0.126/(0.084+0.126) = 0.6

1d)
P(-study | smart, -pass) = P(-study, smart, -pass) / P(smart, -pass) = 0.054/ (0.054 + 0.006) = 0.9

1e)
P(pass|smart) = P(pass, smart) / P(smart)

= (P(pass,smart,-study) + P(pass,smart,study))/P(smart)

= (0.126+0.114)/(0.3)

= 0.8

1f)

P(pass|study) = P(pass, study)/ P(study)

=(P(pass,study,smart) + P(pass, study, -smart))/P(study)

=(0.168+0.114)/0.4

= 0.705

2a)

P(Cold,Sneeze, Allergic, Scatches, Cat) =

P(Cold)P(Sneeze|Cold^ Allergy)P(Allergy|Cat)P(Scratches|Cat)P(Cat)

2b)

P(-cold,sneeze,allergic,scratches,cat) =

P(-cold)P(sneeze|-cold ^ allergy)P(allergy|cat)P(scratches|cat)P(cat) =

0.95 x 0.7 x 0.75 x 0.5 x 0.02 = 0.0049875

2c)

P(cat|-cold,sneeze,allergic,scratches) =

P(cat,-cold,sneeze,allergic,scratches)/(P(cat,-cold,sneeze,allergic,scratches)+P(-cat,-cold,sneeze,allergic,scratches))

P(cat,-cold,sneeze,allergic,scratches)=

P(-cold)P(sneeze|-cold ^ allergy)P(allergy|cat)P(scratches|cat)P(cat) =

0.95 x 0.7 x 0.75 x 0.5 x 0.02 = 0.0049875

P(-cat,-cold,sneeze,allergic,scratches) =

P(-cold)P(sneeze|-cold ^ allergy)P(allergy|-cat)P(scratches|-cat)P(-cat) =

0.95 x 0.7 x 0.05 x 0.05 x 0.98 = 0.00162925
0.0049875/(0.0049875+0.00162925) = 0.75376884422

2d)

P(cat|scratches) = (P(scratches|cat)P(cat))/P(scratches)

Numerator: P(scratches|cat)*P(cat) = 0.5 x 0.02 = 0.01

2e)

Denominator: P(scratches) =

P(scratches ^ cat) + P(scratches ^ -cat) =

P(scratches|cat)P(cat) + P(scratches ^ -cat)P(-cat)

= (0.5x0.02) + (0.05x0.98)

Requires 4 joint probabilities from the JPT.

3a)

To start a car, you have to be at the car and have the key, and the car has to have a charged battery and the tank has to have gas. Afterwards, the car will be running, and you will still be at the car and have the key after starting the engine.

Action (startCar(x, y, z,a, b))
//Have(x,y) means x has y
Precond: operator(x) ^ car(y) ^ key(z) ^ battery(a) ^ tank(b) ^ At(x, y) ^ Have(x,z) ^ Have(y,a) ^ charged(x,a) ^ hasGas(y,b)
Effect: running(y) ^ At(x, y) ^ Have(x, z)

3b)
∀x,y, z,a,b,s operator(x,s) ^ car(y,s) ^ key(z,s) ^ battery(a,s) ^ tank(b,s) ^ At(x,y,s) ^ Have(x,z,s) ^ Have(y,a,s) ^ charged(x,a,s) ^ hasGas(x,b,s) → running(y,do(startCar(x,y,z,a,b),s)) ^ At(x,y,do(startCar(x,y,z,a,b),s)) ^ Have(x,z,do(startCar(x,y,z,a,b),s))

3c) *starting this car will not changes whether any other car is out of gas (tank empty)*

∀x,y, z,a,b,s, c,d operator(x,s) ^ car(y,s) ^ key(z,s) ^ battery(a,s) ^ tank(b,s) ^ At(x,y,s) ^ Have(x,z,s) ^ Have(y,a,s) ^ charged(x,a,s) ^ hasGas(x,b,s) ^ car(c) ^ tank(d) ^ c ≠ y ^ d ≠ b → [hasGas(c,d,s) ↔hasGas(c,d,do(startCar(x,y,z,a,b),s))]